

TOPOLOGY OF METRIC SPACES

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A Self Study

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1 Spaces, Balls and Openness

DEFINITION 1.1 (Metric Space). Let X be a non empty set, and $d : X \times X \rightarrow \mathbb{R}$ be a metric on X such that $\forall x, y, z \in X$:

1. $d(x, y) \geq 0$, and $d(x, y) = 0 \iff x = y$
2. $d(x, y) = d(y, x)$
3. $d(x, z) + d(z, y) \geq d(x, y)$

Then the pair (X, d) is called a *metric space*.

DEFINITION 1.2 (Inner Product Space). If V is a real vector space, and $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is called an inner product if $\forall x, y, z \in V$:

1. $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0 \iff x = 0$
2. $\langle x, y \rangle = \langle y, x \rangle$
3. $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$

$$4. \langle \alpha x, y \rangle = \alpha \langle x, y \rangle \text{ where } \alpha \in \mathbb{R}$$

Then $(V, \langle \cdot, \cdot \rangle)$ is called an *inner product space*.

DEFINITION 1.3 (Norm Linear Space). Let V be a real vector space and $\|x\|$ be the norm of a vector $x \in V$, then $(V, \|\cdot\|)$ is a norm linear space if $\forall x, y \in V$:

$$1. \|x\| \geq 0 \text{ and } \|x\| = 0 \iff x = 0.$$

$$2. \|ax\| = |a| \|x\| \text{ where } a \in \mathbb{R}.$$

$$3. \|x + y\| \leq \|x\| + \|y\|.$$

THEOREM 1.4 (Cauchy–Schwarz Inequality). Let V be an inner product space and $\|x\| = \sqrt{\langle x, x \rangle}$ be the norm of $x \in V$, then for $x, y \in V$ we have:

$$|\langle x, y \rangle| \leq \|x\| \|y\| \quad (1)$$

Proof. First, we assume $\|x\| = \|y\| = 1$, then consider $\langle x - y, x - y \rangle$:

$$\langle x - y, x - y \rangle = \langle x, x \rangle + \langle y, y \rangle - 2 \langle x, y \rangle \quad (2)$$

$$= 2(1 - \langle x, y \rangle) \geq 0 \implies \langle x, y \rangle \leq 1 \quad (3)$$

Similarly, for $\langle x + y, x + y \rangle$ we get $\langle x, y \rangle \geq -1$. Hence,

$$|\langle x, y \rangle| \leq 1 = \|x\| \|y\| \quad (4)$$

Now, for the general non-zero case, let $u = x/\|x\|$ and $v = y/\|y\|$, then $\|u\| = \|v\| = 1$ and $\langle u, v \rangle \leq 1$, by the previous case.

$$\left\langle \frac{x}{\|x\|}, \frac{y}{\|y\|} \right\rangle \leq 1 \implies \langle x, y \rangle \leq \|x\| \|y\| \quad (5)$$

The case for $\|x\| = 0$ or $\|y\| = 0$ are trivial. □

DEFINITION 1.5. Let V be a real vector space and $x, y, z \in V$. Then z lies between x and y $\iff \exists t \in [0, 1]$ such that $z = tx + (1 - t)y$.

THEOREM 1.6. Let $x, y \in V$, where V is an inner product space with the norm metric. Then $d(x, y) = d(x, z) + d(z, y) \iff z$ lies between x and y .

Proof. If z lies between x and y , then $z = tx + (1 - t)y$ for some $t \in [0, 1]$, by the triangle

inequality,

$$d(x, z) + d(z, y) = \|x - z\| + \|z - y\| \quad (6)$$

$$= \|x - tx + (t - 1)y\| + \|tx + (1 - t)y - y\| \quad (7)$$

$$= \|(1 - t)x - (1 - t)y\| + \|tx - ty\| \quad (8)$$

$$= (1 - t)\|x - y\| + t\|x - y\| = \|x - y\| \quad (9)$$

$$= d(x, y) \quad (10)$$

If we assume $d(x, y) = d(x, z) + d(z, y)$, then $\|x - y\| = \|x - z\| + \|z - y\|$, which implies that $x - z$ and $y - z$ are linearly dependent and $x - z = k(z - y)$ where $k \geq 0 \implies z = \frac{1}{1+k}x + \frac{k}{1+k}y$, and let $t = \frac{1}{1+k}$. Since $t \in [0, 1]$, we have $z = tx + (1 - t)y$ i.e. z lies between x and y . $\square$

LEMMA 1.7 (Young's Inequality). *Let x, y be two non negative real numbers. Let $p > 1$ and q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then the following inequality holds:*

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \quad (11)$$

Proof. We fix $y > 0$, and define the following function:

$$f(x) := \frac{x^p}{p} + \frac{y^q}{q} - xy \quad (12)$$

Applying the extrema test of single variable, we get that $f'(x_0) = 0$, and see that it holds for $x_0 = y^{1/(p-1)}$. Then we also see that $f''(y^{1/(p-1)}) > 0$, from which we can conclude that it is a minimum of the function on $(0, \infty)$, from which it follows that $f(x) \geq f(x_0)$, and here $f(x_0)$ happens to be zero. Therefore:

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \quad (13)$$

$\square$

LEMMA 1.8 (Hölder's Inequality). *Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and for $1 \leq p \leq \infty$, we define $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ and for $p \rightarrow \infty$, we have $\|x\|_\infty = \max\{|x_i| : 1 \leq i \leq n\}$. For $p > 1$, let q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then the following inequality holds:*

$$\sum_i |a_i| |b_i| \leq \|a\|_p \|b\|_q \quad \forall a, b \in \mathbb{K}^n \quad (14)$$

Proof. Let us take $x = \frac{|a_i|}{\|a\|_p}$ and $y = \frac{|b_i|}{\|b\|_q}$ and using Young's Inequality, we get:

$$\frac{|a_i| |b_i|}{\|a\|_p \|b\|_q} \leq \frac{1}{p} \left(\frac{|a_i|}{\|a\|_p} \right)^p + \frac{1}{q} \left(\frac{|b_i|}{\|b\|_q} \right)^q \quad (15)$$

Summing this over i from 1 to n , we get:

$$\frac{1}{\|a\|_p \|b\|_q} \sum_{i=1}^n |a_i| |b_i| \leq \frac{1}{p \|a\|_p^p} \sum_{i=1}^n |a_i|^p + \frac{1}{q \|b\|_q^q} \sum_{i=1}^n |b_i|^q \quad (16)$$

Simplifying this, we obtain:

$$\sum_{i=1}^n |a_i| |b_i| \leq \|a\|_p \|b\|_q \quad (17)$$

□

LEMMA 1.9 (Minkowski's Inequality). *Let $1 \leq p \leq \infty$, then we have Minkowski's inequality:*

$$\|a + b\|_p \leq \|a\|_p + \|b\|_p \quad \forall a, b \in \mathbb{K}^n \quad (18)$$

Proof. For $1 < p < \infty$, we observe that:

$$\|a + b\|_p = \left(\sum_{i=1}^n |a_i + b_i|^p \right)^{1/p} \quad (19)$$

$$\leq \left(\sum_{i=1}^n |a_i| |a_i + b_i|^{p-1} + \sum_{i=1}^n |b_i| |a_i + b_i|^{p-1} \right) \quad (20)$$

Now, we apply Hölder's inequality to both the terms, by taking $p = q(p-1)$ and $1/q = (p-1)/p$. Applying it to the left term:

$$\sum_{i=1}^n |a_i| |a_i + b_i|^{p-1} \leq \|a\|_p \left[\sum_{i=1}^n (|a_i + b_i|^{p-1})^q \right]^{1/q} \quad (21)$$

$$= \|a\|_p \left[\sum_{i=1}^n |a_i + b_i|^p \right]^{1/q} \quad (22)$$

$$= \|a\|_p \|a + b\|_p^{p/q} \quad (23)$$

Similarly for the right term, we have

$$\sum_{i=1}^n |b_i| |a_i + b_i|^{p-1} \leq \|b\|_p \|a + b\|_p^{p/q} \quad (24)$$

Using these two relations and simplifying, we finally obtain:

$$\|a + b\|_p \leq \|a\|_p + \|b\|_p \quad (25)$$

□

DEFINITION 1.10 (Open and Closed Balls). Let (X, d) be a metric space, and $x \in X$. Then for

some $r > 0$, we define the open ball $B(x, r)$ and closed ball $B[x, r]$ as follows:

$$B(x, r) := \{y \in X \mid d(x, y) < r\} \quad (26)$$

$$B[x, r] := \{y \in X \mid d(x, y) \leq r\} \quad (27)$$

DEFINITION 1.11 (Line Segment). Let V be a real vector space and $x, y \in V$. We denote the line segment joining x and y by $[x, y]$, and we define to be $[x, y] := \{tx + (1 - t)y \mid 0 \leq t \leq 1\}$.

DEFINITION 1.12 (Convex Set). A subset $A \subset V$ of a vector space is said to be *convex* if $\forall x, y \in V$, the line segment $[x, y] \subset A$.

DEFINITION 1.13 (Open Set). For some $U \subset (X, d)$, the subset is said to be *open* if $\exists r > 0$ such that $B(x, r) \subset U \forall x \in U$.

LEMMA 1.14. Let (X, d) be a metric space, $x \in X$ and $r > 0$. Then the open ball $B(x, r)$ is an open set.

Proof. Let $y \in B(x, r)$. Assuming that $\exists s > 0$ such that $B(y, s) \subset B(x, r)$, if $z \in B(y, s)$, we want to show that $d(z, x) < r$. Now,

$$d(z, x) \leq d(z, y) + d(y, x) \leq s + d(x, y) < r$$

Therefore, we have $d(z, x) < r$, and thus $B(y, s) \subset B(x, r)$. Since y is arbitrary, we have proved the open ball $B(x, r)$ is an open set. $\square$

DEFINITION 1.15 (Interval). If $I \subseteq \mathbb{R}$, then I is said to be an interval if and only if any $x, y \in I$ such that $x < y$ and $\forall z \in \mathbb{R}$ such that $x < z < y \implies z \in I$.

PROPOSITION 1.16. Any non empty subset $I \subseteq \mathbb{R}$ is an interval if it is of the form (a, b) , $[a, b]$, $(a, b]$, $[a, b)$, $(-\infty, \beta)$, (α, ∞) , $(-\infty, \beta]$ or $[\alpha, \infty)$, where $b < a$.

Proof. Dividing this into two cases: (1) bounded set and (2) unbounded set.

(1) Let $\sup I = b$ and $\inf I = a$. We claim that $(a, b) \subset I \subset [a, b]$. For $a < x < b$, then $x > a$ since a is the infimum, hence $\exists c \in I$ such that $c < x$. Similarly $x < b \implies \exists d \in I$ such that $x < d$, thus $c < x < d$ and it follows that $x \in I$ since its an interval. Since $x \in (a, b)$ is arbitrary, therefore $(a, b) \subset I$. As such, if we have $z \in I$ then $a \leq z \leq b$ by their definitions and hence $I \subset [a, b]$. Depending whether a and b actually are in I , we get (a, b) , $[a, b]$, $(a, b]$, $[a, b)$.

(2) For the unbounded case, let $b = \sup I$ and we claim $(-\infty, b) \subset I \subset (-\infty, b]$. Since b is a supremum, for some $x \in I$ we have $x < b$ and therefore $\exists d \in I$ such that $x < d$. We can

also always find $c \in I$ such that $c < x$ and then we have $c < x < d \implies x \in I$. Therefore $(-\infty, b) \subset I \subset (-\infty, b]$, and we get the interval based on whether b is in I or not. Similarly, one can proceed with the (a, ∞) and $[a, \infty)$ cases. $\square$

LEMMA 1.17. *A non empty open set in $\mathbb{R}$ is the union of countable family of pairwise disjoint open intervals.*

Proof. Let $U \subseteq \mathbb{R}$ be an open set and $x \in U$. Now let I_x be the union of all sets such that $x \in I \subset U$, where I is an interval. Then, I_x would be the maximal interval containing x in U . Now, for $x, y \in U$, consider the intervals I_x and I_y . If they are not disjoint i.e. $I_x \cap I_y \neq \emptyset$, then their union would be a larger interval containing both, which would no longer be the maximal set unless $I_x = I_y$. So I_x and I_y must be either disjoint or identical. Then, we can write U can be written as $\bigcup_{i \in I} I_i$, where I is the index set.

Now, to prove countability, choose a rational $r_i \in I_i$ for each $i \in I$. Then, $\forall i \neq j \implies r_i \neq r_j$ since $I_i \cap I_j = \emptyset$, so we have a one to one map $i \mapsto r_i$ from I to $\mathbb{Q}$, hence I is countable. $\square$

THEOREM 1.18. *Let (X, d) be a metric space and let $\mathcal{T}$ denote the set of all open sets in X , then:*

1. $\emptyset, X \in \mathcal{T}$.
2. Arbitrary union of sets in $\mathcal{T}$ lies in $\mathcal{T}$.
3. Arbitrary intersection of sets in $\mathcal{T}$ lies in $\mathcal{T}$.

Proof.

1. $\emptyset$ is open by definition, and obviously X is open, so both are in $\mathcal{T}$.
2. Let $\{U_i \mid i \in I\}$ be a family of open sets. If U_k is open, then $\exists r > 0$ such that $B(x, r) \subset U_k \subset \bigcup_{i \in I} U_i$. Since x is arbitrary, it follows that the union of open sets is open in X , and it belongs to $\mathcal{T}$.
3. Let U_k be a finite collection of open sets in X . If the collection is pairwise disjoint, then $\bigcap_{k=1}^n U_k = \emptyset$ which is open in X . If not, let $x \in \bigcap_{k=1}^n U_k$, and since U_k is open then $\exists r_k > 0$ such that $B(x, r_k) \subset U_k$. Let $r := \min\{r_k \mid 1 \leq k \leq n\}$ and it follows that $B(x, r) \subset B(x, r_k) \subset U_k$ for each k i.e. $B(x, r) \subset \bigcap_{k=1}^n U_k$.

$\square$

DEFINITION 1.19 (Topological Space). Let X be a non empty set, then if $\mathcal{T}$ is a collection of subsets of X such that:

1. $\emptyset, X \in \mathcal{T}$.

2. An arbitrary union of sets in $\mathcal{T}$ lies in $\mathcal{T}$.
3. An arbitrary intersection of sets in $\mathcal{T}$ lies in $\mathcal{T}$.

Then $\mathcal{T}$ is said to be a *topology* of X and $(X, \mathcal{T})$ is called a *topological space*.

DEFINITION 1.20 (Equivalent Metrics). Two metrics d and d' are said to be *equivalent* if the topologies generated by the two are the same. This is equivalent to saying that $U \subset X$ is d -open if and only if it is d' -open.

DEFINITION 1.21 (Interior). Let $S \subset X$ be a subset of a metric space. If $x \in S$, it is said to be an *interior point* of S if $\exists r > 0$ such that $B(x, r) \subset S$. The set of interior points of S is denoted by S^0 and is called the *interior* of the set S .

If $(X, \mathcal{T})$ is a topological space, and $x \in A \subset X$ is said to be an interior point if $\exists$ an open set U with $x \in U$ such that $U \subset A$.

DEFINITION 1.22. Let $Y \subset X$ of a metric space. A subset $A \subset Y$ is said to be open in Y if it is an open subset of metric space (Y, d) where d is the induced metric on Y .

DEFINITION 1.23 (Subspace Topology). If $Y \subset X$ is a subset of a topological space $(X, \mathcal{T})$, then the *subspace topology* $\mathcal{T}_Y$ is the collection $\{U \cap Y \mid U \in \mathcal{T}\}$.

DEFINITION 1.24 (Closed Set). A subset $F \subset X$ of a metric space is said to be *closed* if its complement $X - F$ is open in X .

THEOREM 1.25. Let $\mathcal{C}$ denote the class of closed sets in a metric space X , then:

1. $\emptyset, X \in \mathcal{C}$.
2. The intersection of an arbitrary family of closed sets is closed.
3. The union of a finite arbitrary family of closed sets is closed.

Proof.

1. $\emptyset$ and X are closed by definition.
2. Let $\{F_i \mid i \in I\}$ be a family of closed sets in $\mathcal{C}$. Since F_i is closed, its complement $U_i := X - F_i$ is open. From Theorem 1.18, we know the union of all such open sets is open and hence its complement, the intersection of all closed sets is closed.
3. Let $\{F_i\}$ be a finite set of closed subsets of X . Then the complement of F_i is U_i must be open, and from Theorem 1.18 we know a finite intersection of such sets is open. We take its complement to get that the finite union of closed is closed.

□

2 Convergence

DEFINITION 2.1 (Sequence). A sequence in a (metric) space X is a function $x : \mathbb{N} \rightarrow X$. We exhibit the sequence x as (x_n) where $x_n := x(n)$ for $n \in \mathbb{N}$. A subsequence is the restriction of the map to $x : S \rightarrow X$ where $S \subseteq \mathbb{N}$. If we exhibit $n_1 < n_2 < \dots < n_k$ then we write the subsequence as (x_{n_k}) .

LEMMA 2.2. For a subsequence (x_{n_k}) of a sequence (x_n) , $n_k \geq k$.

Proof. We know that $n_1 \geq 1$, then since $n_2 > n_1$ we have $n_2 \geq 2$, and assume $n_k \geq k$, then $n_{k+1} \geq k+1$ since $n_{k+1} > n_k$, and by induction, this is true for all $k \in \mathbb{N}$. □

DEFINITION 2.3 (Convergence). A sequence (x_n) converges to $x \in X$ if $\forall \epsilon > 0 \exists N \in \mathbb{N}$ such that $x_n \in B(x, \epsilon)$, or in other words $d(x, x_n) < \epsilon$. Then we say the sequence is convergent and the sequence has the limit x . It is written as $x_n \rightarrow x$ or $x = \lim_{n \rightarrow \infty} x_n$.

On an arbitrary topological space, the sequence (x_n) converges to $x \in X$ if for all open sets $U \ni x \exists N$ such that for $n \geq N$ we have $x_n \in U$.

LEMMA 2.4. The limit of a sequence in a metric space is unique.

Proof.

Method 1: Let $x_n \rightarrow x$ and $x_n \rightarrow y$. Then we have the following:

$$\begin{aligned} \forall \epsilon > 0 \exists N_1 \in \mathbb{N} \text{ such that } n \geq N_1 &\implies d(x, x_n) < \epsilon \\ \forall \epsilon > 0 \exists N_2 \in \mathbb{N} \text{ such that } n \geq N_2 &\implies d(y, x_n) < \epsilon \end{aligned}$$

Now, choose $N = \max\{N_1, N_2\}$, then for $n \geq N$ we have $d(x, y) \leq d(x, x_n) + d(x_n, y) < 2\epsilon$. Since for arbitrary $\epsilon > 0$ we have $d(x, y) < 2\epsilon$, then $d(x, y) = 0 \implies x = y$. □

Method 2: We use the Hausdorff property, if $x \neq y$ then $\exists r > 0$ such that $B(x, r) \cap B(y, r) = \emptyset$. Now, let $\delta := d(x, y)$ and choose $r \leq \delta/2$ and look at the open balls $B(x, r)$ and $B(y, r)$, and we claim they are disjoint. If so, then for $z \in B(x, r) \cap B(y, r)$,

$$\delta = d(x, y) \leq d(x, z) + d(z, y) < r + r \leq \delta,$$

which is a contradiction. Now, since $x_n \rightarrow x$ then $\exists n_1$ such that for $n \geq n_1$ we have $x_n \in B(x, r)$ and similarly $\exists n_2$ such that for $n \geq n_2$ we have $x_n \in B(y, r)$, $\forall r > 0$. In particular, we see for $n \geq \max\{n_1, n_2\}$ you see $x_n \in B(x, r) \cap B(y, r)$. This contradicts that the balls are disjoint, and hence $x = y$. □

In topological spaces, this lemma does not hold, except for a specific class of topological spaces called Hausdorff spaces, which we define below.

DEFINITION 2.5 (Hausdorff Space). A topological space X is said to be *Hausdorff* if $\forall$ distinct $x, y \in X$, one can find open sets $U \ni x$ and $V \ni y$ such that $U \cap V = \emptyset$.

DEFINITION 2.6 (Limit Point). Let X be a metric space and let $E \subset X$. A point $x \in X$ is called a *limit point* of E iff $\forall r > 0$, we have $B(x, r) \cap E \neq \emptyset$.

In a topological space, $x \in X$ is a limit point of E iff $\forall$ open sets $U \ni x$, we have $U \cap E \neq \emptyset$.

THEOREM 2.7. A subset E of a metric space X contains all its limit points iff E is closed in X .

Proof. ($\implies$) Assume that E contains all its limit points i.e. $\forall x \in \tilde{E} \exists r > 0$ such that $B(x, r) \cap E = \emptyset$. This means that the open ball is completely contained outside of E , hence $B(x, r) \subset \tilde{E}$. Therefore, $\forall x \in \tilde{E} \exists r > 0$ such that $B(x, r) \subset \tilde{E}$. Hence $\tilde{E}$ is open and its complement E is closed.

($\impliedby$) Assume E is a closed set, i.e. its complement $\tilde{E}$ is open. Hence, $\forall x \in \tilde{E} \exists r > 0$ such that $B(x, r) \subset \tilde{E}$. Now, let y be a limit point of E in X such that $y \in \tilde{E}$, then by the open nature of $\tilde{E}$ we have an $s > 0$ such that $B(y, s) \subset \tilde{E}$, then $B(y, s) \cap E = \emptyset$. This contradicts the fact that y is a limit point, hence the only limit points of E are contained in E . $\square$

COROLLARY 2.8. A subset E of a metric space X is closed iff (x_n) is a sequence in E converging to $x \in X$, then $x \in E$.

DEFINITION 2.9 (Cluster Point). A point x is a *cluster* or *accumulation* point of $E \subset X$ if $\forall r > 0$, the open ball $B(x, r)$ contains a point other than x .

If we let $B'(x, r) := B(x, r) \setminus \{x\}$, then $B'(x, r)$ is called a *deleted neighbourhood* of x . Then, x is a cluster point iff every deleted neighbourhood of x contains a point of E .

THEOREM 2.10 (Bolzano–Weierstrass). Let A be an infinite bounded subset of $\mathbb{R}$, then there is a cluster point of A in $\mathbb{R}$.

Proof. Let $E := \{x \in \mathbb{R} \mid x \leq a \text{ for infinitely many } a \in A\}$. Let $M \in \mathbb{R}$ be such that $-M \leq a \leq M \forall a \in A$. It is clear that $-M \in E$ and M is an upper bound of E . Then, $\exists \ell \in \mathbb{R}$ such that $\sup(E) = \ell$. We claim ℓ is a cluster point of A i.e. $\forall \epsilon > 0 \exists a \in (\ell - \epsilon, \ell + \epsilon) \cap A$ such that $a \neq \ell$. Since ℓ is the supremum of E , then $\forall \epsilon > 0 \exists x \in E$ such that $x > \ell - \epsilon$. Then, we have $a > \ell - \epsilon$. Also, for infinitely many a , we have $a < \ell + \epsilon$. Otherwise, for finitely many such elements of A for all other $a \in A$, we have $a \geq \ell + \epsilon$, then $\ell + \epsilon \in E$ which contradicts $\ell = \sup(E)$. Then, this implies that $\exists$ infinitely many $a \in A$ such that $\ell - \epsilon < a < \ell + \epsilon$ i.e. $\exists a \in A \cap (\ell - \epsilon, \ell + \epsilon)$ which is different from ℓ . $\square$

COROLLARY 2.11. Any bounded sequence (x_n) on $\mathbb{R}$ has a convergent subsequence (x_{n_k}) .

Proof. Let (x_n) be a bounded sequence on $\mathbb{R}$, then $\exists x \in \mathbb{R}$ such that $x = x_n$ for each $n \in \mathbb{N}$. Consider a subsequence with indices (n_k) , let x be a cluster point of the image of (x_{n_k}) . Let $x_{n_k} \in (x - 1/k, x + 1/k)$ be an element of the sequence chosen inductively such that $x_{n_{k+1}} \in \{x_{n_1}, x_{n_2}, \dots, x_{n_k}\}$. Then the subsequence (x_{n_k}) converges to x . $\square$

DEFINITION 2.12 (Bounded Set). We define A to be *bounded* in (X, d) if $\exists x_0 \in X$ and $R > 0$ such that $A \subset B(x_0, R)$.

DEFINITION 2.13 (Cauchy Sequence). A sequence (x_n) in a metric space X is said to be *Cauchy* iff $\forall \epsilon > 0 \exists N \in \mathbb{N}$ such that for any $m \geq N$ and $n \geq N$, we have $d(x_m, x_n) < \epsilon$.

PROPOSITION 2.14. Let (X, d) be a metric space. Then the following are true:

1. Any convergent sequence in X is also Cauchy.
2. Any Cauchy sequence in X is bounded i.e. $\exists x \in X$ and $r > 0$ such that $x_n \in B(x, r) \forall n \in \mathbb{N}$.
3. A Cauchy sequence is convergent if it has a convergent subsequence.

Proof.

1. Let $x_n \rightarrow x$ be a convergent sequence in X . Then $\forall \epsilon > 0 \exists N \in \mathbb{N}$ such that for $n \geq N$, we have $d(x_n, x) < \epsilon$. Now, for some $m \geq N$, we have $d(x_m, x) < \epsilon$. Then, by triangle inequality we write $d(x_n, x_m) \leq d(x_n, x) + d(x_m, x)$, then we have $d(x_n, x_m) < 2\epsilon$. Since ϵ is arbitrary, we get $\forall \epsilon > 0 \exists N \in \mathbb{N}$ such that for any $m \geq N$ and $n \geq N$, we have $d(x_m, x_n) < \epsilon$, hence the sequence is Cauchy. $\square$
2. Given a Cauchy sequence (x_n) in X , then $\forall \epsilon > 0 \exists N \in \mathbb{N}$ such that for any $m \geq N$ and $n \geq N$, we have $d(x_m, x_n) < \epsilon$. If we set $m = N$, we get $d(x_N, x_n) < \epsilon$ for all $n \geq N$, so it is bounded by the choice of ϵ . Now, for $n < N$, we choose $R := \max\{d(x_i, x_N) \mid 1 \leq i < N\}$, then $d(x_n, x_N) < R$ for each $n < N$. For each ϵ and R , we now choose $r := \max\{\epsilon, R\}$, then $\forall n \in \mathbb{N}$, we have $x_n \in B(x_N, r)$. $\square$
3. Let (x_n) be a Cauchy sequence such that its subsequence (x_{n_k}) converges to x . Then, for a given $\epsilon > 0 \exists N \in \mathbb{N}$ such that for $n, m \geq N$, then $d(x_n, x_m) \leq \epsilon$. Now, this is true for any $n_k \geq N$ as well, then $d(x_n, x_{n_k}) < \epsilon$. Using the convergence of the subsequence, we find $K \in \mathbb{N}$ such that for $k \geq K$, we have $d(x_{n_k}, x) < \epsilon$. Choosing $\eta := \max\{n_K, N\}$, we next use $d(x, x_n) \leq d(x_{n_k}, x_n) + d(x, x_{n_k})$ then for $n, n_k \geq \eta$, we get $d(x, x_n) < 2\epsilon$. Hence, the Cauchy sequence (x_n) converges if there exists a subsequence that converges. $\square$

DEFINITION 2.15 (Complete Space). A metric space (X, d) is said to be *complete* iff every Cauchy sequence converges to point in X .

THEOREM 2.16 (Completeness of $\mathbb{R}$). The set of real numbers $\mathbb{R}$ is complete with respect to the metric $d(x, y) = |x - y|$ for all $x, y \in \mathbb{R}$.

Proof. Consider a Cauchy sequence (x_n) on $\mathbb{R}$. We know that every Cauchy sequence is bounded, but we also know every bounded sequence on $\mathbb{R}$ has a convergent subsequence. We also know that every Cauchy sequence with a convergent subsequence is convergent. Hence, every Cauchy sequence on $\mathbb{R}$ is a convergent sequence. $\square$

THEOREM 2.17 (Completeness of $\mathbb{R}^n$). The set $\mathbb{R}^n$ is complete with respect to the metric $d(x, y) = \|x - y\|$ for all $x, y \in \mathbb{R}^n$.

Proof. Consider a Cauchy sequence (x_m) on $\mathbb{R}^n$ such that $x_m = (x_{m1}, x_{m2}, \dots, x_{mn})$. We know that if the sequence (x_{mi}) converges for $1 \leq i \leq n$, then (x_m) converges. First, we claim that (x_{mi}) is a Cauchy sequence if (x_m) is Cauchy. Given that $\forall \epsilon > 0 \exists N \in \mathbb{N}$ such that for any $m \geq N$ and $k \geq N$, we have $\|x_m - x_k\| < \epsilon$. Now, we know that $|x_{mi} - x_{ki}| \leq \|x_m - x_k\|$, then $|x_{mi} - x_{ki}| < \epsilon$, and hence (x_{mi}) is Cauchy. Since (x_{mi}) is a sequence on $\mathbb{R}$, then it must be convergent by completeness, and since each coordinate sequence is convergent, then (x_m) must be convergent on $\mathbb{R}^n$ if it is Cauchy. $\square$

DEFINITION 2.18 (Diameter). Let A be a non empty subset of (X, d) , then the *diameter* $\text{diam}(A)$ is defined by

$$\text{diam}(A) := \sup\{d(x, y) \mid x, y \in A\}$$

DEFINITION 2.19 (Dense Set). We say that a subset D of (X, d) is *dense* in X if $\forall x \in X$ and $r > 0$, we have $B(x, r) \cap D \neq \emptyset$.

In a topological space, a subset $D \subset X$ is dense if for every non empty open set $U \subset X$, we have $D \cap U \neq \emptyset$.

LEMMA 2.20. Let $S := \{n + m\sqrt{2} \mid n, m \in \mathbb{Z}\}$. Let $a, b \in \mathbb{R}$ be such that $a < b$. Then $\exists s \in S$ such that $a < s < b$. In other words, S is dense in $\mathbb{R}$.

Proof. im lazy, so just check the book. $\square$

DEFINITION 2.21 (Basis). Let X be a (metric) space, a family of open sets $\mathcal{B} := \{B_i \mid i \in I\}$ indexed by indexing set I is a *basis* for the topology on X iff any open set in X is the union of some collection of members of $\mathcal{B}$.

DEFINITION 2.22 (Boundary). Let X be a (metric) space and $A \subset X$. A point $x \in X$ is said to be a *boundary* point of A in X if every open set $U \ni x$ obeys $A \cap U \neq \emptyset$ and $\bar{A} \cap U \neq \emptyset$. The set of all boundary points is called the *boundary* of A , denoted by ∂A .

3 Continuity